The Effects of Menopause on the Belly Button Microbiome

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Introduction

The microbiome of the belly button has been the least explored, compared to other microbiomes. The exploration of the belly button can lead us to novel discoveries and new understandings of how hormones affect the belly button. Menopause is a time where hormones are in flux, and as a result, can radically change the environment that the microorganisms in the belly button inhabit.

In this study, we analyzed the amount of *Staphylococcus* present in the belly button from women aged 18-32, the fertile range, and woman aged 49+, the average age for menopause (NCBI).

Research Question

We examined the microbiome of women in menopause (49+) and fertile women (18-32) to determine if menopause would cause the amount of the bacteria *Staphylococcus* to decrease due to the decrease of hormones.

Methods

1. Using Phinch 2.0 software, set the appropriate values and export the data
2. Analyzed the data for women ages 18-32 (n=43) and 49+ (n=32) examining the 5 most common sequences in each age set
3. Compare how much *Staphylococcus* has decreased or increased

Results

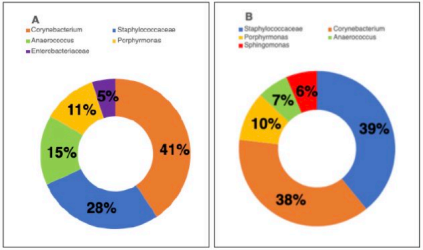

Figure 1. Characteristics of the microbiome of women in different ages. Graphs representing the five most common genera in the belly button microbiome of: (A) women aged 49 and above and (B) women aged 18-32.

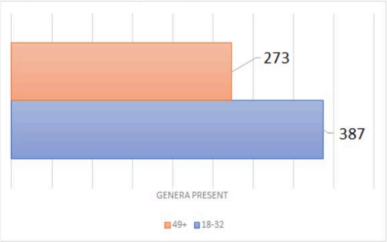

Figure 2. Aging-related changes in the diversity of women's belly button microbiomes. Graph representing the total number of genera sequences for each age group (49+ and 18-32)

References

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Discussion

- The hypothesis was supported by our data, as the amount of the bacteria *Staphylococcus* decreased
- *Corynebacterium* was the most common in women ages 49+.
- The abundance of each top 5 bacteria varied from before and after menopause
- There is significantly less genera present in women 49+.
- It is possible that the environment created without the hormone created an ideal living
- It is possible that there are other environmental influence that affects the data.

Future Directions

- The data set was limited in that analyzed data set was collected by a previous study (Hulcr et al., 2012)
- Implement more specific subdivisions (age group, stages of menopause, before first period, etc)

Figure 3. Schematic Diagram of factors that could affect the bacteria in the belly button. The factors above are important aspects of what affects the kind of that could be considered in a future study.

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